

Sample Test 4
Times: 90 minutes

Question 1 (2 points). Let

$$\phi \equiv \forall x \exists y (P(x, y) \wedge \neg Q(x, y, z))$$

and

$$\psi \equiv \exists x (P(x) \wedge Q(f(x, y))) \rightarrow \forall z R(z, y).$$

- a) Draw parse trees and identify the free and bound variables for ϕ and ψ .
- b) Determine $\phi[h(z)/z]$ and $\psi[g(x)/y]$.

Question 2 (2 points). Verify the following sequents using natural deduction rules.

- a) $\forall x P(x) \vee \forall x Q(x) \vdash \forall x (P(x) \vee Q(x))$
- b) $\neg \forall x \neg P(x) \vdash \exists x P(x)$

Question 3 (2 points). For each of the formulas below, find a model that satisfies the formula and one that does not.

- a) $\forall x \forall y ((P(x) \rightarrow P(y)) \wedge (P(y) \rightarrow P(x)))$
- b) $\forall x \exists y (P(x) \rightarrow Q(y)) \rightarrow (\forall x P(x) \rightarrow \exists y Q(y))$

Question 4 (2 points). Which of the following semantic entailments hold?

- a) $\forall x \exists y P(x, y) \models \exists y \forall x P(x, y)$
- b) $\exists x (P(x) \wedge Q(x)) \models \exists x P(x) \wedge \exists x Q(x)$

Question 5 (2 points). Use the tableau algorithm to determine whether the following formulas are valid or not. If it is not valid, is it satisfiable?

- a) $\phi \equiv ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
- b) $\psi \equiv \exists x (P(x) \rightarrow Q(x)) \rightarrow (\forall x P(x) \rightarrow \exists x Q(x))$